

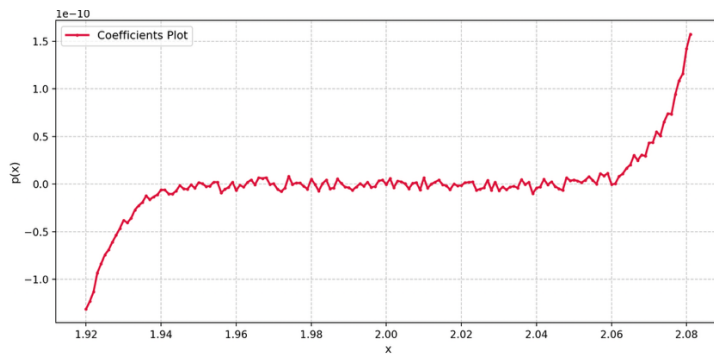
APPM 4600

Homework 1

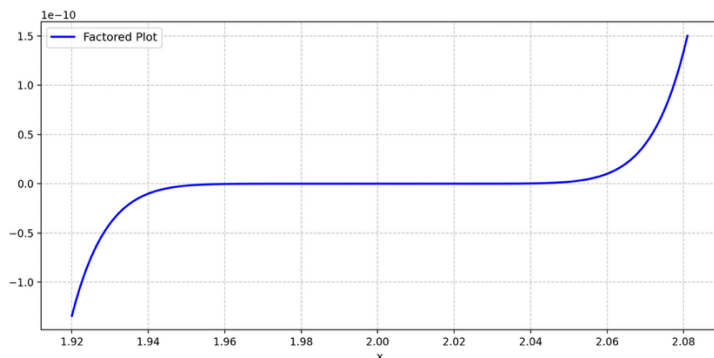
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1.

i) Coefficients Plot: $p(x) = x^9 - 18x^8 + 144x^7 - 672x^6 + 2016x^5 - 4032x^4 + 5376x^3 - 4608x^2 + 2304x - 512$



ii) Factored Plot: $p(x) = (x-2)^9$



iii) The factored plot is correct due to catastrophic cancelling loss in the coefficients plot.

2.

i. Evaluate $\sqrt{x+1}-1$ for $x \approx 0$.

Multiply and divide by $(\sqrt{x+1}+1)$ gives $x/((\sqrt{x+1}+1))$ which avoids subtracting near identical terms

ii. Evaluate $\sin(x) - \sin(y)$ for $x \approx y$.

$\sin(x) - \sin(y) = 2\sin((x-y)/2)\cos((x+y)/2)$ moves the subtraction inside

iii. Evaluate $1 - \cos(x)$ for $x \approx 0$.

$1 - \cos(x) = 2\sin^2(x/2)$ which avoids $1-1$ since $\cos(0)=1$

3.

The second degree Taylor polynomial $P_2(x)$ for $f(x) = (1 + x + x^3) \cos(x)$ about $x_0 = 0$.

$$P_2(x) = 1 + x - (1/2)x^2$$

i. Use $P_2(0.5)$ to approximate $f(0.5)$.

$$P_2(.5) = 1 + .5 - (.5)(.5^2) = 1.375$$

$$f_2(.5) = (1 + .5 + .5^3)(\cos(.5)) = 1.625(\cos(.5)) \approx 1.42607$$

with error 0.05107

$$|f(x) - P_2(x)| \leq (|f'''(\xi)|/3!)|x - x_0|^3$$

$$f'''(x) = -18x\sin(x) - 3(3x^2 + 1)\cos(x) + (x^3 + x + 1)\sin(x) + 6\cos(x)$$

$$f'''(0) = -3(1)(1) + 6 = 3$$

$$\text{for } M=3.01, |f(x) - P_2(x)| \leq (3.01/3!)|x - x_0|^3 = .0627 > 0.05107$$

ii. Find a bound for the error $|f(x) - P_2(x)|$ when $P_2(x)$ is used to approximate $f(x)$. This will be a function of x .

$$|f(x) - P_2(x)| \leq (M/3!)|x - x_0|^3 = (M/6)|x|^3$$

$$f'''(x) = -18x\sin(x) - 3(3x^2 + 1)\cos(x) + (x^3 + x + 1)\sin(x) + 6\cos(x)$$

$$|f(x) - P_2(x)| \leq (3.01/6)|x|^3 \approx .502|x|^3$$

iii. Approximate $\int_0^1 f(x)dx$ using $\int_0^1 P_2(x)dx$

$$\int_0^1 f(x)dx \approx \int_0^1 P_2(x)dx = \int_0^1 (1+x-(1/2)x^2)dx = [x + (1/2)x^2 - 1/6x^3]_0^1 = (1+(1/2)(1^2)-(1/6)(1^3))$$

$$\int_0^1 f(x)dx \approx \int_0^1 P_2(x)dx = 4/3 \approx 1.333$$

iv. Estimate the error

from $f(x) = 1 + x - (1/2)x^2 + (1/2)x^3$ the first neglected term is $((f'''(0))/3!)x^3 = (1/2)x^3$

$$\text{Error} \approx \int_0^1 (1/2)x^3 dx = [(1/8)x^4]_0^1 = 1/8 = .125$$

4. Consider the quadratic equation $ax^2 + bx + c = 0$ with $a=1, b=-56, c=1$.

i. compute the relative error for the roots from the quadratic formula

$$f(x) = x^2 - 56x + 1$$

$$(-(-56) + \sqrt{(-56)^2 - 4(1)(1)})/(2(1)) = (-56 + \sqrt{3132})/2 = (-56 + 55.964)/2 = 55.982$$

$$(-(-56) - \sqrt{(-56)^2 - 4(1)(1)})/(2(1)) = (-56 - \sqrt{3132})/2 = (-56 - 55.964)/2 = 0.018$$

The true roots are 55.98213718 and 0.01786284

$$|(55.98213718 - 55.982)/55.98213718| \approx 2.45 \times 10^{-6}$$

$$|(0.01786284 - 0.018)/0.01786284| \approx 7.68 \times 10^{-3}$$

ii.

$$(x-r_1)(x-r_2) = x^2 - (r_1+r_2)x + r_1r_2 = 0 \text{ with } x^2 + (b/a)x + (c/a) = 0$$

$$(r_1+r_2) = -(b/a) \text{ gives } r_2 = -(b/a) - r_1$$

$(r_1r_2) = c/a$ gives $r_1 = c/(ar_2)$ where $r_1 = c/(ar_2)$ avoids the subtraction that increased the error in the second root